Lecture 5

January 22, 2026

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Key Drivers of Portfolio Credit Risk

- Probability of default (PD)
  - physical (real-world) default rates
- Loss given default (LGD)
  - seniority, collateral, covenants
- Correlation
  - defaults and downgrades cluster in downturns
- Concentration
  - issuer, sector, geography, capital structure

Unexpected Loss

- Expected Loss is the average credit loss over the horizon - the long-run cost of holding credit risk
  - priced into spreads
  - provisioned and budgeted
- Unexpected Loss is the variability of losses around EL
- Arises because:
  - defaults are rare but severe
  - defaults cluster in downturns
  - recoveries fall in stress
- Drives solvency risk, not pricing

Why Unexpected Loss Matters

- Two portfolios can have:
  - identical expected loss
  - very different tail risk
- Unexpected loss determines:
  - capital requirements
  - risk limits
  - portfolio construction decisions

Expected Loss vs Unexpected Loss

- Expected Loss
  - average outcome
  - priced and provisioned
- Unexpected Loss
  - tail outcomes
  - managed via capital and limits

Credit Value-at-Risk (Credit VaR)

- Credit VaR measures a loss threshold over a given horizon
- Definition:
  - Loss not exceeded with probability α
- Typical choices:
  - horizon: 1 year
  - confidence: 99% or 99.9%
- How bad could portfolio credit losses be in a severe but plausible scenario?
- Captures default clustering and concentration effects, but does not describe average outcomes

Expected Shortfall (Conditional VaR)

- Expected Shortfall (ES), also called Conditional VaR, measures:

$$ES_\alpha = \mathbb{E}[\text{Loss} \mid \text{Loss} \geq VaR_\alpha]$$

  - Loss = portfolio loss (non-negative)
  - α = confidence level (e.g. 99%)
  - VaRα = α-quantile of the loss distribution
- Average loss in the worst 1−α% of scenarios, so:

$$ES_\alpha = \frac{1}{1-\alpha}\int_\alpha^1 \mathrm{VaR}_u \, du$$

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In the quantile representation, VaR_u denotes VaR evaluated at probability level u; Expected Shortfall is the average of these VaRs over the tail $u \in [\alpha, 1]$.

Why Expected Shortfall Is Preferred in Credit

- Credit loss distributions are:
  - Skewed
  - fat-tailed
- ES captures:
  - multiple defaults
  - low recoveries
  - systemic stress
- Better aligned with capital and solvency risk

Portfolio Credit Risk Tools

- Historical default and recovery data
- Scenario analysis and stress testing
- Portfolio credit models to:
  - simulate joint outcomes
  - translate credit events into losses

Credit VaR vs. Expected Shortfall

- Credit Value-at-Risk (VaR)
  - Loss threshold at confidence level α
  - Think of it as losses will not exceed this level with probability α.
  - Captures severity at a cutoff
  - However, ignores what happens beyond the threshold
- Expected Shortfall
  - Average loss given losses exceed VaR
  - Think of it as how, in the worst $1-\alpha\%$ of scenarios, large the losses are on average.
  - Captures tail severity

CreditMetrics

- CreditMetrics is a framework for assessing portfolio credit risk arising from changes in the market value of debt due to changes in obligor credit quality.
- Developed by J.P. Morgan's Risk Management Research division (1997)
- Firms are not running the original CreditMetrics implementation, but the framework provides the intuition behind most modern credit portfolio risk models.

Model Philosophy and Inputs

- Model philosophy:
  - Seeks to balance multiple sources of information rather than relying solely on recent market moves
  - Uses broad historical data to capture full credit cycles
  - Models the full range of credit quality migration — upgrades, downgrades, and default — not just default events
- Model inputs:
  - Accepts any default risk assessment that classifies obligors into discrete credit states with associated default probabilities
  - Examples include:
    - Agency credit ratings
    - Internal rating systems
    - Statistical default prediction models

Why Credit Portfolio Risk Is Hard

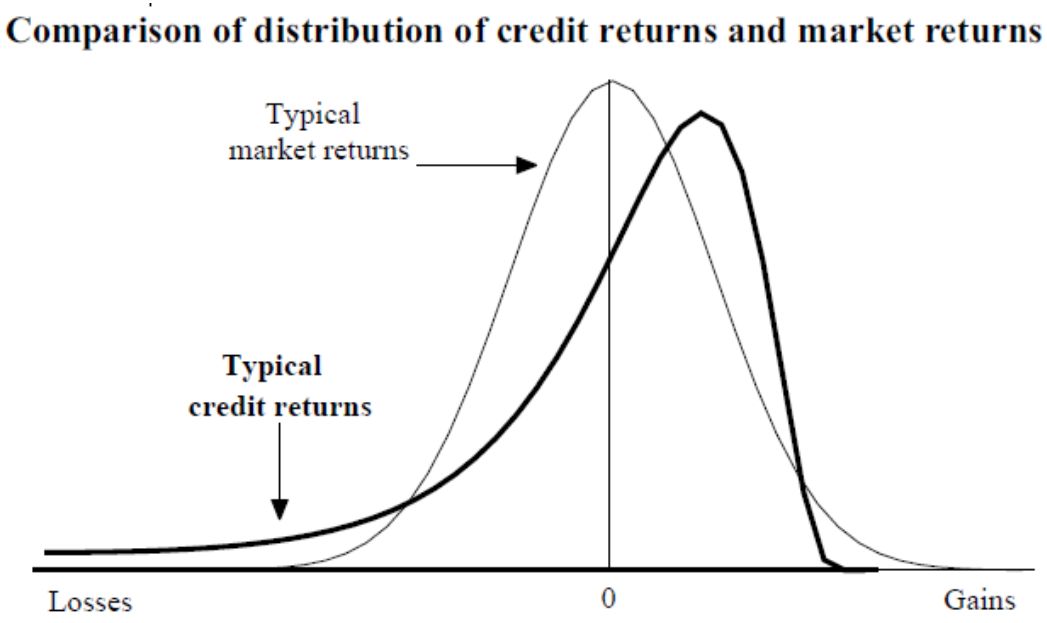
- Applying standard portfolio theory to credit portfolios is neither analytically nor practically straightforward
- While modern portfolio theory works well for equity price risk, fundamental differences make it problematic for credit risk
- Two core challenges distinguish credit portfolios from equity portfolios:
  - Distributional issues
  - Correlation issues

Problem 1: Non-Gaussian Credit Returns

- Equity returns are relatively symmetric and well approximated by a Gaussian (normal) distribution
- In a Gaussian setting, mean and standard deviation fully characterize risk, including percentile outcomes
- Credit returns, by contrast, are highly skewed and fat-tailed, so mean and volatility are insufficient
- Understanding credit portfolio risk therefore requires analysis of the entire loss distribution, not just its first two moments

Credit Returns vs Market Returns

- Credit return distributions exhibit a long downside tail driven by rare but severe default losses
- A high likelihood of small interest-driven gains combined with a low probability of large default losses produces a smooth, strongly downside-skewed distribution



Source: CreditMetrics Technical Document, RiskMetrics Group (2007).

Problem 2: Modeling Credit Correlation

- Portfolio credit risk depends on joint credit quality changes, not isolated defaults
- Unlike equities, credit correlations cannot be directly estimated from high-frequency market prices
- Limited default and migration data make empirical estimation difficult
- Practical models therefore rely on simplifying assumptions or latent-factor frameworks to capture correlation

CreditMetrics: How the Framework Addresses the Two Problems

- Addresses non-Gaussian credit losses by using historical (physical) credit migration and default probabilities to model discrete credit state transitions (upgrades, downgrades, default), translating those transitions into changes in instrument value and producing a full, asymmetric portfolio loss distribution
- Addresses correlation explicitly by modeling credit quality transitions jointly across obligors, typically through correlated latent firm-value shocks, so that downgrades and defaults cluster in adverse states of the world
- By aggregating revalued positions across joint scenarios, CreditMetrics produces expected loss and tail-risk measures that reflect diversification benefits and concentration risk

CreditMetrics: Simplified Portfolio Framework

- Step 1: Portfolio Inputs
  - Exposures (bonds / loans)
  - Credit states (ratings or internal grades)
  - Seniority and recovery assumptions
  - Credit spreads
- Step 2: Credit State Transitions
  - Rating migration probabilities:
    - Upgrades
    - Downgrades
    - default
  - Correlated credit quality changes across obligors

CreditMetrics: Simplified Portfolio Framework (con.)

- Step 3: Revaluation of Positions
  - Each credit state implies a different spread
  - Migration → mark-to-market repricing
  - Default → recovery-based valuation
- Step 4: Portfolio Loss Distribution
  - Aggregate revalued positions
  - Construct portfolio loss distribution
  - Measure:
    - Expected Loss (EL)
    - Credit VaR
    - Expected Shortfall (ES)

CreditMetrics: Overall Modeling Steps

- CreditMetrics decomposes portfolio credit risk analysis into a sequence of logical steps:
  A. Compute credit risk at the individual bond level
  B. Aggregate risk across bonds to form a portfolio loss distribution
  C. Analyze how correlation and diversification affect tail risk
  D. Translate tail risk into economic capital
  E. Attribute portfolio risk back to individual exposures

See: Malhotra, Yogesh. CreditMetrics Methodology and Credit Value at Risk (Credit VaR)

A) Credit VaR Computation for Each Bond

- CreditMetrics begins at the single-instrument level, focusing on how credit events affect bond value over a fixed horizon.
- The objective is to derive a distribution of forward bond values, not just an expected loss.
- Step 1: Specify the Credit Risk Horizon
  - Choose a fixed horizon over which credit quality may change
  - Common choice: 1 year, consistent with rating transition data and Credit VaR definitions
  - All credit events (migration or default) are assumed to occur within this horizon
  - This defines when credit risk is measured.

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A) Step 2: Specify the Transition Matrix

- Use a rating transition matrix to describe possible credit quality outcomes over the horizon
- Transition probabilities are based on historical (physical) data
- Default is treated as a terminal credit state
- This defines what credit states are possible and their likelihoods.

Illustrative Example Historical Rating Paths

ObligorID	Year	Year	Year	Year	Year	Year	Year	Year	Year	Year	Year1	Year1	Year1	Year1
OBL0001	AA	AA	AA	D	D	D	D	D	D	D	D	D	D	D
OBL0002	AAA	AAA	AAA	AAA	AAA	AA	AA	AA	AA	AA	AA	AA	AA	AA
OBL0003	A	A	A	A	A	A	A	A	A	A	A	A	BBB	BBB
OBL0004	B	B	BB	BB	BB	BB	D	D	D	D	D	D	D	D
OBL0005	AA	AA	AA	AA	D	D	D	D	D	D	D	D	D	D
OBL0006	BBB	BBB	BBB	BBB	BBB	BBB	BBB	BBB	BBB	BBB	BBB	BBB	BBB	BBB
OBL0007	A	A	A	A	BBB	BBB	BBB	BB	BB	BB	BB	BB	D	D
OBL0008	BB	BB	BB	BB	BB	BB	BB	BB	BB	BB	BB	BB	BB	B
OBL0009	BB	BB	BB	BB	BB	BB	BB	BB	BB	BB	BB	B	B	B
OBL0010	BB	BB	B	B	B	B	B	CCC	CCC	BB	BB	CCC	CCC	CCC
OBL0011	BB	A	A	A	A	A	A	A	BB	BB	BB	BB	CCC	CCC
OBL0012	BBB	BBB	BBB	BBB	BBB	BBB	BBB	BBB	BBB	BB	BB	BB	BB	BB
OBL0013	BBB	BB	BB	BBB	BBB	BBB	BBB	BBB	BBB	BBB	A	D	D	D
OBL0014	A	A	A	A	A	A	D	D	D	D	D	D	D	D
OBL0015	A	A	BBB	BBB	BB	BB	BB	BB	BB	BB	BB	BB	BBB	BBB
OBL0016	AAA	AAA	AAA	AAA	AAA	AAA	AAA	AAA	AAA	AAA	AAA	AAA	AAA	AAA
OBL0017	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA	A	A	A

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Annual Rating Transition Matrix

From \ To	AAA	AA	A	BBB	BB	B	CCC	D
AAA	91.00%	7.50%	0.50%	0.00%	0.00%	0.00%	0.00%	1.00%
AA	2.00%	88.00%	6.00%	1.00%	1.00%	0.00%	0.00%	2.00%
A	0.00%	4.00%	87.00%	5.00%	2.00%	0.00%	0.00%	2.00%
BBB	0.00%	1.00%	5.00%	82.00%	6.00%	3.00%	0.00%	3.00%
BB	0.00%	0.00%	2.00%	4.00%	80.00%	8.00%	4.00%	3.00%
B	0.00%	0.00%	0.00%	1.50%	6.00%	78.50%	9.00%	5.00%
CCC	0.00%	0.00%	0.00%	0.00%	3.00%	7.00%	72.00%	18.00%
D	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%

A) Step 3: Specify the Forward Pricing Model

- Assign a forward value to the bond under each possible end-of-horizon credit state
- Forward valuation reflects:
  - remaining cash flows
  - credit spreads associated with the new rating
  - recovery value in default
  - Credit losses arise from repricing, not only default
- This defines how credit events translate into value changes.

A) Step 4: Derive the Forward Distribution of Bond Value

- Combine transition probabilities and forward values by credit state
- Produce a discrete probability distribution of bond value at the horizon
- This distribution is typically:
  - Asymmetric
  - fat-tailed
  - dominated by downside risk

- This completes Credit VaR computation for a single bond.

(B) Portfolio Credit VaR Computation

- Portfolio credit risk is computed by combining joint credit outcomes across all bonds
- Each portfolio outcome consists of:
  - a portfolio value (sum of bond values under that joint outcome)
  - an associated probability, derived from the joint likelihood of individual credit states
- The collection of all joint outcomes and probabilities forms the portfolio loss distribution
- Portfolio risk emerges from the probability-weighted aggregation of joint credit states, not from adding individual risks.

(C) Correlation and Diversification Analysis

- Credit quality changes across obligors are not independent
- CreditMetrics introduces correlation by modeling joint credit state transitions
- Correlation determines:
  - default clustering
  - tail losses
  - limits to diversification benefits
- This is where portfolio theory truly enters credit risk.

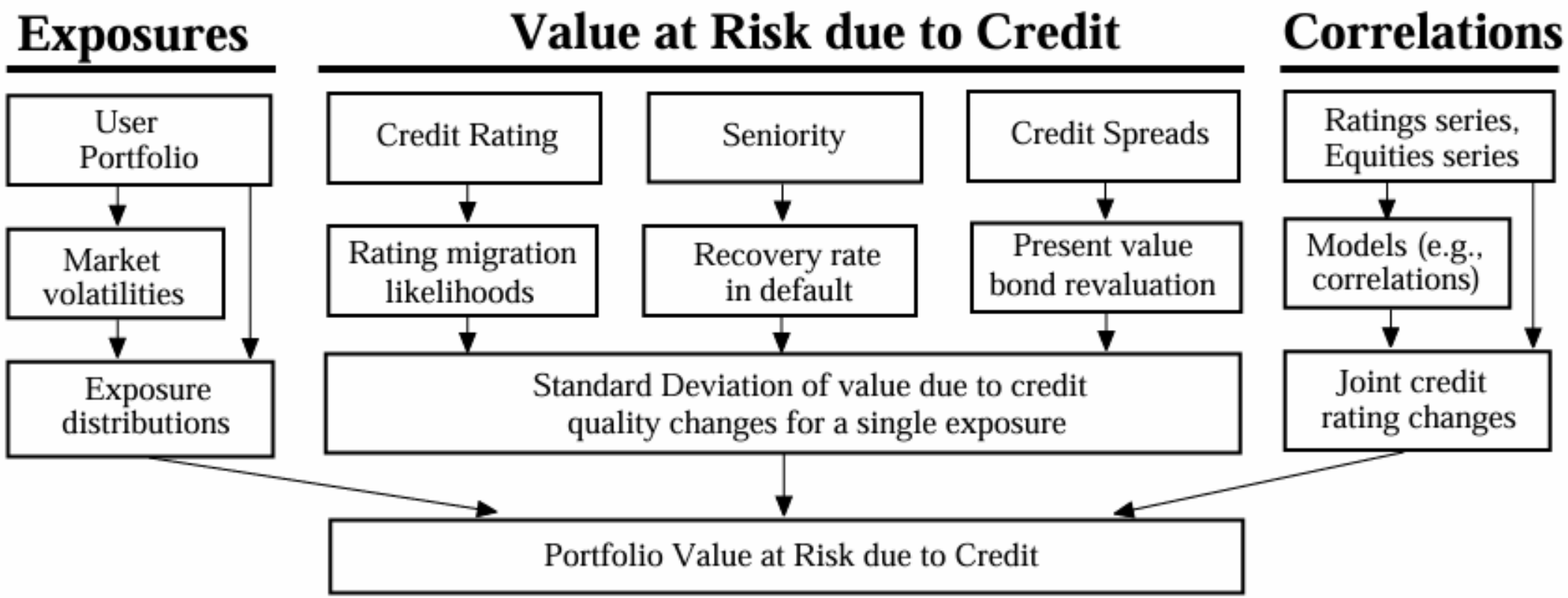
(D) Credit VaR and Economic Capital

- From the portfolio loss distribution, can compute:
  - Expected Loss (EL)
  - Credit Value-at-Risk (VaR)
  - Expected Shortfall
- Economic capital is set to absorb unexpected losses beyond EL
- Capital reflects solvency risk, not pricing

(E) Marginal Risk Contribution and Diversification Benefits

- Portfolio Credit VaR can be decomposed into marginal contributions by bond
- Marginal risk depends on:
  - Size
  - credit quality
  - correlation with the rest of the portfolio
- Enables analysis of:
  - diversification benefits
  - concentration penalties
  - risk-adjusted portfolio construction

CreditMetrics Framework



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